

RESEARCH ARTICLE

Bare soil cover and arbuscular mycorrhizal community in the first montane forest restoration in Central Argentina

Alejandra G. Becerra^{1,2} , Adriana Diván¹, Daniel Renison³

Soil erosion affects extensive areas worldwide and must be urgently reduced promoting plant cover and beneficial microorganisms associated with plants, including arbuscular mycorrhizal fungi (AMF). In mountain environments, plant cover is difficult to enhance due to harsh conditions during the dry season and steep slopes. Our objective was to evaluate the percentage of the soil surface covered by plants and the AMF community associated with trees 12.5 years after planting during forest restoration efforts in microsites at different levels of soil degradation. The study was performed in the first montane forest restoration initiative of Central Argentina, where one of the trials consisted of planting *Polylepis australis* saplings at microsites with different levels of soil degradation: high, intermediate, and low. After 12.5 years, percentage of bare soil cover was significantly reduced by 36 and 37% in the high and intermediate degradation microsites, respectively. Low degradation microsites were initially very low in bare soil and did not significantly change. Mycorrhizal colonization, hyphae, vesicles, arbuscules, AMF diversity, and community structure were similar among microsite types. Percentage of hyphal entry points was higher at microsites with low degradation, number of spores was higher in high and intermediate degradation, and species richness was higher in high degradation. *Acaulospora* and *Glomus* were the most abundant genera in all microsites. We conclude that even in the most degraded microsites around 2.8% of the bare soil is covered by vegetation each year and that the arbuscular mycorrhizal community is highly tolerant and adapted to soils with different disturbance types.

Key words: arbuscular mycorrhizal community, bare soil cover, mountain forest, restoration

Implications for Practice

- Bare soil cover was reduced by excluding livestock and planting native trees even in the most degraded microsites (outside of erosion gullies).
- The arbuscular mycorrhizal fungi community was resilient to degradation and all its species were present even in the most degraded microsites.

Introduction

Soil erosion and loss due to agriculture and livestock production activities affect extensive areas worldwide, causing severe environmental, social, and economic damage (Pimentel et al. 1995). Soil loss reduces the organic matter content of the remaining soils, the soil capacity to absorb rainfall, the soil moisture, and the soil-dwelling organisms (FAO 1993; Doetterl et al. 2016). Even when the factors triggering erosion processes are removed, sites which lost all or most of the plant cover would be particularly susceptible to slow or null recovery for plant colonization (Green et al. 2005; Doetterl et al. 2016). Promoting vegetation system recovery in the affected areas in order to achieve a suitable cover that minimizes continuous soil loss requires intervention actions that are costly to implement, especially in mountain areas where the steep slopes and low temperatures predispose

soil loss and reduce plant growth (Körner 2003; Cingolani et al. 2013).

Bare soils need to be protected with plant cover in order to reduce soil loss, reduce water runoff, augment soil stability, and soil-dwelling organisms (Durán Zuazo & Rodríguez Pleguezuelo 2008; FAO 2015). An important component of soil-dwelling organisms is mycorrhizal fungi, which build extensive mycelial networks associated with the great majority of forest plants, and mediate soil formation and structure (Leifheit et al. 2014; Meier et al. 2019). Among the mycorrhizal types, arbuscular mycorrhizal fungi (AMF) are associated with almost 80% of the roots of vascular plants (Wang & Qiu 2006). AMF facilitate plant mineral nutrition, mostly of phosphorus and other macro- and micronutrients, nitrogen, zinc, copper, and magnesium (Clark & Zeto 2000; Smith et al. 2000), protect

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¹Instituto Multidisciplinario de Biología Vegetal (IMBIV), CONICET, Universidad Nacional de Córdoba, Casilla de Correo 495, 5000 Córdoba, Argentina

²Address correspondence to A. Becerra, email abecerra@unc.edu.ar

³Instituto de Investigaciones Biológicas y Tecnológicas, Centro de Ecología y Recursos Naturales Renovables (IIByT-CERNAR), CONICET, Universidad Nacional de Córdoba, Vélez Sarsfield 1611, 5000 Córdoba, Argentina

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plants from root pathogens (Sikes 2010), and help plants tolerate drought (Augé 2001). In degraded soils, fertility is usually low and plant growth is limited compared to less degraded soils, while the presence of AMF could enhance growth and development of their host plants (Smith & Read 2008; Fajardo et al. 2011).

In the mountains of Central Argentina, extensive deforestation (Zak et al. 2004) and soil erosion processes might transform this region into a rock desert (Cingolani et al. 2013). With the aim of reversing this trend and restoring native forest cover, the first Argentine forest restoration project was started in 1997 and more projects followed in subsequent years (Renison et al. 2016). Here, the native tree *Polylepis australis* BITT (Rosaceae) is one of the most abundant trees and one of the most widely used species in restoration of upland forests. This tree is considered a keystone species providing habitat for numerous taxa including birds and wood decaying fungi, and promoting the regeneration of shade-tolerant plant species (Robledo & Renison 2010; Renison et al. 2011; Bellis et al. 2015). The AMF associated with *P. australis* are highly tolerant and adapted to soils with different disturbance types: high livestock stocking density (Menoyo et al. 2009), reduced forest structural complexity (Soteras et al. 2015), or fires (Soteras et al. 2013). However, no evaluation of the AMF associated with *P. australis* has included highly degraded soils, which have remained bare of vegetation for decades.

Our aim was to evaluate change in bare soil cover and the AMF community associated with *P. australis* saplings 12.5 years after planted in an area where livestock was excluded and which included microsites with different degrees of degradation. The levels of degradation were categorized according to the percentage of the bare soils covered by vegetation and by the amount of exposed mineral soils, which lost the top organic layer (Renison et al. 2005). Two questions are addressed: (1) Does livestock exclusion and tree planting reduce the percentage of bare soils under all degrees of degradation? and (2) How is the AMF community affected by degree of initial soil degradation? We propose a worst-case scenario hypothesis where forest restoration actions would promote a reduction of bare soil at microsites with intermediate degradation but not at the most degraded microsites, and that the AMF community associated with planted *P. australis* would be negatively affected by the characteristics of the high degradation microsites.

Methods

Study Area

Los Gigantes massif is located in the north of the high mountains of Central Argentina (31°24'46"S, 64°48'22"W, 2,300 m a.s.l.). The area was historically used mainly for livestock rearing and firewood extraction for local use. In recent years, the area has also been used for trekking, running, climbing, and other mountain sports. The mean temperature at 2,100 m a.s.l. in the coldest and warmest months is 5 and 11.4°C, respectively (Cabido 1985). Mean annual precipitation averaged 900 mm from 1992 to 2010, with 83% of the rainfall occurring during

the warm months between October and April. Soils in the area are mostly Mollisols composed of fine-textured eolian deposits and material derived from the weathering of the local granite substrate. Soils are dark, with high organic matter content, and according to the U.S.D.A Soil Taxonomy may be classified as Cumulic Hapludoll, Entic Hapludoll, and Aquic Argiudoll at upper, mid, and low slopes (Cabido et al. 1987; Soil Survey Staff 2010).

The landscape consists of a mosaic of tussock grasslands, grazing lawns, granite outcrops, eroded areas with exposed soil and rock surfaces partially colonized by annuals, mosses and lichens, and woodlands of *Polylepis australis* that cover about 12% of the area (Fig. 1A & 1B; Cingolani et al. 2004; Coladon et al. 2010). Other less abundant woody species present in the study area include the tree *Maytenus boaria* Molina (Celastraceae), the shrubs *Escallonia cordobensis* (Kuntze) Hosseus (Glossulariaceae), *Berberis hieronimi* C.K. Schneid. (Berberidaceae), and *Satureja* spp. (Lamiaceae), and the dwarf shrub *Gaultheria poeppigii* DC. (Ericaceae) (Cingolani et al. 2014).

The study site is within a 22 ha livestock enclosure built as part of the first forest restoration project of Central Argentina initiated in 1997 and still ongoing. According to local informants, the area lost all its forests in 1920 and now around 30% of the surface consists of bare rock, which in the past was covered by soils (Cingolani et al. 2004). Restoration work consisted of planting 35,000 saplings mainly of *P. australis*, but also *M. boaria* and *E. cordobensis*, and the ongoing work of reducing soil erosion in 26 gullies by manually smoothing soil and planting with native vegetation (Renison et al. 2016). The site is very heterogeneous with slopes ranging from 0 to 90° and varied topographies including deep ravines, open valleys, slopes, ridges, and rock outcrops of varying sizes, which provided a varying amount of protection from human-related activities such as livestock trampling, fires, and trekking at a small scale. As a consequence of the interaction of topography and human activities, vegetation cover was also very variable at the star of the restoration project and bare soils were found at erosion gullies.

Sampling Design

To guide our sampling design, we used information from 655 circular plots 60 cm in diameter categorized in the year 1998 as of high, intermediate, and low degradation (Renison et al. 2005). These plots at planting had a bare soil cover of 81, 45, and 5.7% and a remaining top layer of organic soil cover of 0.1, 78, and 91%, respectively, with the underlying mineral soil being easily distinguished due to its reddish hue. Soil depth and slope were relatively similar between degradation types with a mean depth of 76 cm and a slope of 10° (in Renison et al. 2005: microsites 1.1, 2.2, and 3.2). We avoided choosing plots within the steep edges of erosion gullies where it was not possible to plant the trees, thus our findings apply to areas out of erosion gullies.

All these plots were marked in 1998 in the field with a numbered metal pin and a *P. australis* sapling was planted at the plot center (age of 6–7 months, height of 3.9 ± 0.1 cm). Trees

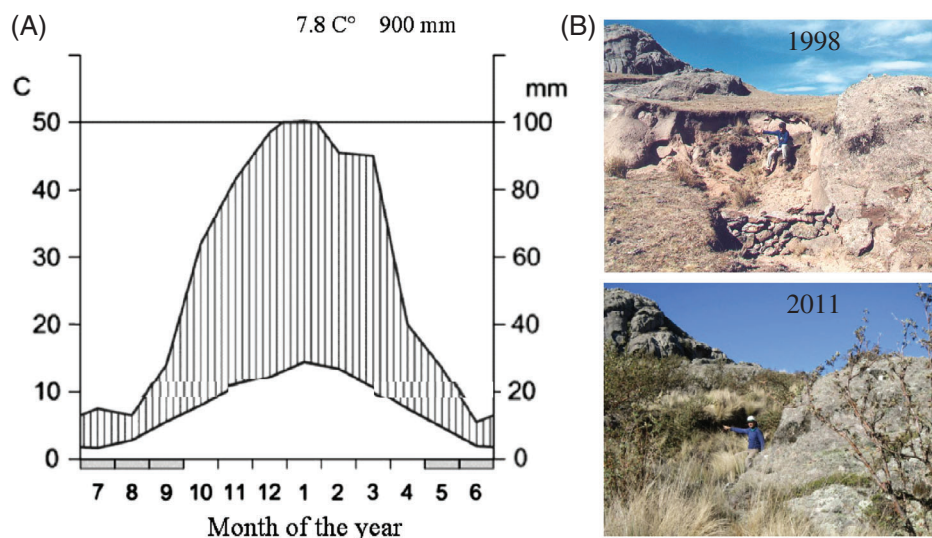


Figure 1. (A) Climatic diagram based on data extracted from WorldClim for the study area at of 2,300 m a.s.l. (Hijmans et al. 2005). (B) Photos in 1998 and 2011 of a section of the study plantations where the heterogeneous nature of the site is visualized. Note the combination of bare soils, mixed, and complete vegetation cover at a small scale in the 1998 photo.

were watered only at planting, no fertilizers were used, and they were monitored for survival and growth thereafter (Renison et al. 2005; Simoes Macayo & Renison 2015).

In 2011, when trees were 13 years old and had been planted for 12.5 years, we selected 12 replicates per type of microsite, totaling 36 plots. To avoid incorporating variability in height of sampled individuals, we preselected planted trees that were closest to the average height for each microsite type and then selected using a random number table. Since the montane forests of Central Argentina have been severely degraded by deforestation and livestock grazing, we could not find a nondisturbed area with similar characteristics to our study restoration site so no control or reference area was included in this study.

Bare Soil

We used the respective estimations of bare soils performed in 1998 when trees were planted and reevaluated bare soil 12.5 years later in 2011. Bare soil was visually estimated as the proportion of surface soil not covered by herbaceous vegetation or dry leaf matter as described by Renison et al. (2005). We did not include the planted tree as vegetation cover because bare soil under the tree canopy is prone to soil erosion (Renison 2011, Universidad Nacional de Córdoba, personal observation).

AMF: Colonization and Spores

For the first time in 2011, when trees had been planted for 12.5 years, we evaluated *P. australis* root colonization and soil spore content in the 36 selected plots. From each individual, root and rhizosphere soil samples were extracted, during autumn, at a distance of between 15 and 50 cm from the main stem, using a soil core 15 cm in diameter $\times$ 15 cm in depth. Samples were placed in plastic bags and stored at 4°C. In the

laboratory, soil samples were placed in trays containing water. *Polylepis australis* roots were separated from the soil and fixed in 70% alcohol. In order to assess the percentage of arbuscular mycorrhizal colonization (AMC), roots were cleared with 10% KOH (15 minutes at 90°C), acidified with 1% HCL (1 minute, room temperature), and stained in 0.05% aniline blue, following the techniques described by Phillips and Hayman (1970) and Grace and Stribley (1991). For the determination of the percentage of total AMC and of intraradical structure colonization (hyphal entry points, arbuscules, vesicles, and hyphae), five permanent slides per sample were mounted and 100 line intersections were counted under a compound microscope at 400 $\times$ magnification, using the magnified intersection method of McGonigle et al. (1990). To determine spore morphospecies and density, soil samples were sieved (1-cm mesh size) to remove litter, stones, and sticks. Then, 100 g of dry soil from each tree was filtered through different mesh size sieves (125 and 38 μ m) (Gerdemann & Nicolson 1963) and centrifuged in 80% sucrose (Walker et al. 1982). Only apparently healthy spores were isolated by direct observation with a stereomicroscope. For quantification and taxonomic identification, fungal spores were mounted onto slides using polyvinyl alcohol (PVL) with and without Melzer's reagent (Omar et al. 1979) and examined under a compound microscope (ECLIPSE E200, NIKON. Nikon Corporation, Tokyo, Japan). AMF morphospecies identification was based on current species classification (Redecker et al. 2013) and the identification manuals of Schenck and Pérez (1987), Blaszkowski AMF site (2013, www.agro.ar.szccecin.pl/jblaszkowski/), and INVAM (2018, <http://fungi.invam.wvu.edu/the-fungi/species-descriptions.html>). Total spore density was determined as the AMF spore number per 100 g of soil dry weight. Species richness was estimated as the number of different species found in all samples and species diversity was calculated using the Shannon–Weaver index (Magurran 1988).

Soil Characterization

Soil samples collected 12.5 years after the trees were planted were evaluated for total C, organic matter (Nelson & Sommers 1982), total N (Bremner & Mulvaney 1982), C:N, soil texture, pH (Peech 1965), electrical conductivity (Bower & Wilcox 1965), extractable P (Jackson 1964), nitrates, and gravels.

Data Analysis

Residual data distribution was normal (modified Shapiro–Wilk normality test) and variances were distributed homogeneously (Levene test), except for number of AMF spores. To compare the effect of the different microsite types on the changes in bare soil from planting date to 12.5 years later, a repeated measures analysis of variance (ANOVA) was conducted with a model including main effects and the interaction of time $\times$ microsite. For explanatory reasons, changes in bare soil were calculated as bare soil at planting (%) minus bare soil 12.5 years later (%). With respect to AMF no initial data are available, consequently all variables associated with fungal parameters represent values 12.5 years after planting. Simple ANOVA were conducted to compare the effect of the different microsite types on percentage of AMC and intraradical fungal structures, number of AMF spores, species diversity, richness, and the relevant community composition axis as summarized using detrended correspondence analysis (DCA; Ter Braak 1987). For the DCA, we used the 30 AMF morphospecies with two or more non-zero records. To identify groups of means that did not overlap, the test of Di Rienzo, Guzmán, and Casanoves (DGC test) was applied in all cases. A Kruskal–Wallis rank sum test was applied to number of AMF spores per genus and post hoc pairwise comparisons were made using the Conover test. All the analyses were performed using InfoStat software (InfoStat 2017).

Results

Bare Soil

Percentage of bare soil when trees were planted was 81, 45, and 5.7% for high, intermediate, and low degradation microsites, respectively, with a reduction of 12.5 years after planting of 36, 37, and 2%, respectively. This reduction in bare soil was significant for the high and intermediate types of microsites, while in the low degradation microsites the reduction in bare soil was not significant (Fig. 2, repeated measures ANOVA: main effects $F = 37.73$, $p < 0.0001$; time $\times$ microsite type $F = 8.21$, $p < 0.001$).

Arbuscular Mycorrhizal Fungi

Percentage of AMC in *Polylepis australis* roots as evaluated 12.5 years after planting ranged between 11.4 and 70% (mean and standard deviation 48.6 ± 13.5) and did not differ between microsites (Table 1). AMC ranged between 20.5 and 70% at high degradation microsites, between 29.8 and 65.6% at the intermediate degradation microsites, and between 11.4 and

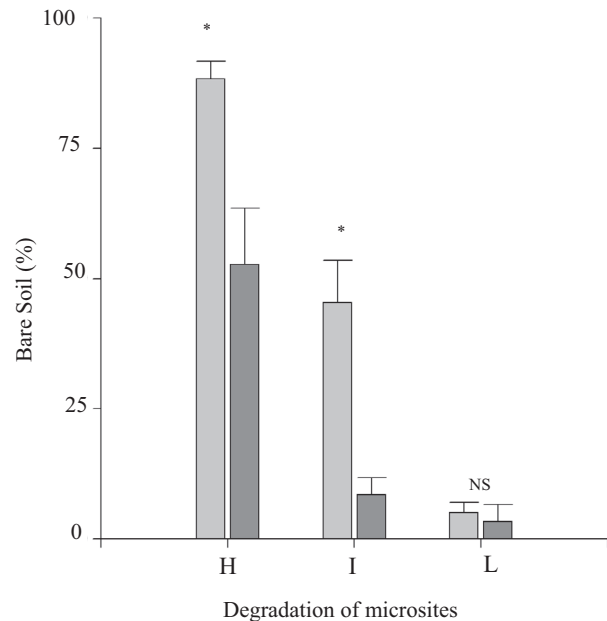


Figure 2. Bare soil percentage in 1998 (year of initiation of restoration, □) and 2011 (year of study, ■) at microsites with different degradation levels (H, high; I, intermediate; L, low). Asterisk “*” indicates significant differences ($p < 0.001$) between years, NS stands for not significant. Error bars represent + SE.

64% at the low degradation microsites. The intraradical fungal structures analyzed were similar among microsites, except for hyphal entry points, which were significantly higher at the low degradation microsites than at the other microsites (Table 1). The number of spores ranged between 141 and 2,068 per 100 g of dry weight of soil (mean and standard deviation 804.5 ± 478.4). The highest number of spores was recorded at high and intermediate degradation microsites (Table 1). Morphospecies richness was significantly higher at the high degradation microsites (Table 1) as compared to the other microsite types, ranging between 7 and 13 (9.6 ± 2.4). Richness varied between 6 and 11 at the intermediate degradation microsites and between 5 and 15 at the low degradation microsites. Shannon–Weaver diversity index did not vary among microsite types (Table 1). A total of 33 AMF morphospecies were identified, of which 20 were attributed to known species belonging to the genera *Acaulospora*, *Ambispora*, *Claroideoglossum*, *Entrophospora*, *Funnelformis*, *Gigaspora*, *Glomus*, *Rhizophagus*, and *Scutellospora* (Table 2). Differences among genera were observed for each microsite type (Table 3). *Acaulospora* spp. and *Glomus* spp. were the most abundant genera in the three microsite types. No significant differences were observed among microsite types for each genus ($p > 0.05$). Confirming these results, at the genus level, the arbuscular mycorrhizal community composition as summarized in three DCA axes with eigenvalues of 0.21, 0.09, and 0.04 did not significantly differ between microsite types (ANOVA: axis 1, $F = 0.18$, $p = 0.84$; axis 2, $F = 0.95$, $p = 0.40$; axis 3, $F = 0.05$,

Table 1. Arbuscular mycorrhizal colonization, intraradical fungal structures (hyphae, arbuscules, vesicles, and entry points), number of arbuscular mycorrhizal fungi spores, richness, and diversity in three types of microsites. Different letters indicate significant differences between microsites ($p < 0.05$). Means and standard error of the variables are shown by microsite. *Per 100 g of dry soil.

	Degradation of the Microsite		
	High	Intermediate	Low
AMC (%)	48.9 $\pm$ 4.7a	52.2 $\pm$ 2.9a	44.8 $\pm$ 3.9a
Hyphae (%)	25.7 $\pm$ 2.8a	25.9 $\pm$ 1.1a	24.2 $\pm$ 1.8a
Arbuscules (%)	0.6 $\pm$ 0.3a	0.83 $\pm$ 0.2a	1.6 $\pm$ 1.1a
Vesicles (%)	20.4 $\pm$ 3.2a	24.4 $\pm$ 2.5a	16.3 $\pm$ 2.5a
Entry points (%)	1.9 $\pm$ 0.6b	1.1 $\pm$ 0.3b	2.9 $\pm$ 0.5a
Number of AMF spores*	881.9 $\pm$ 125.4a	1,010.8 $\pm$ 159.8a	520.6 $\pm$ 87.6b
AMF richness	10.9 $\pm$ 0.5a	8.7 $\pm$ 0.5b	9.2 $\pm$ 0.8b
AMF diversity	2.6 $\pm$ 0.4a	2.4 $\pm$ 0.5a	2.6 $\pm$ 0.3a

Table 2. Occurrence (+) of arbuscular mycorrhizal fungi morphospecies in the rhizosphere of three *Polylepis australis* types of microsites with high (H), intermediate (I), and low (L) degradation.

	Degradation of the Microsite		
	H	I	L
Family Ambisporaceae			
<i>Ambispora leptoticha</i> Walker, Vestberg, and Schuessler	+		+
Family Acaulosporaceae			
<i>Acaulospora bireticulata</i> Rothwell and Trappe	+		+
<i>A. excavata</i> Ingleby and Walker	+	+	+
<i>A. foveata</i> Trappe and Janos		+	+
<i>A. laevis</i> Gerd. and Trappe	+	+	+
<i>A. mellea</i> Spain and Schenck	+	+	+
<i>A. rehmi</i> Sieverd. and Toro			+
<i>A. scrobiculata</i> Trappe	+	+	+
<i>A. tuberculata</i> Janos and Trappe	+	+	+
<i>Acaulospora</i> sp. 1	+	+	+
<i>Acaulospora</i> sp. 2	+	+	+
<i>Acaulospora</i> sp. 3	+		
<i>Acaulospora</i> sp. 4	+		+
<i>Acaulospora</i> sp. 5	+	+	
Family Gigasporaceae			
<i>Gigaspora margarita</i> Becker and Hall	+	+	+
<i>Gigaspora</i> sp. 1	+		
<i>Scutellospora biornata</i> Spain, Sieverd. and Toro	+	+	+
<i>S. reticulata</i> Walker and Sanders	+	+	
<i>Scutellospora</i> sp. 1	+		
Family Claroideoglomeraceae			
<i>Claroideoglossum claroideum</i> Walker and Schuessler	+	+	+
Family Glomaceae			
<i>Funneliformis geosporum</i> Walker and Schuessler	+	+	+
<i>F. mosseae</i> Walker and Schuessler	+		+
<i>Glomus brohultii</i> Herrera, Ferrer, and Sieverd.	+	+	+
<i>G. fuegianum</i> Trappe and Gerd.	+	+	+
<i>Glomus</i> sp. 1			+
<i>Glomus</i> sp. 2	+	+	+
<i>Glomus</i> sp. 3	+		+
<i>Glomus</i> sp. 4	+	+	+
<i>Glomus</i> sp. 5	+		+
<i>Glomus</i> sp. 6			+
<i>Rhizophagus clarus</i> Walker and Schuessler	+	+	+
<i>R. intraradices</i> Walker and Schuessler	+		+
Insertae sedis			
<i>Entrophospora infrequens</i> Ames and Schneid.	+		+
Total number of morphospecies	29	19	28

Table 3. Genera of arbuscular mycorrhizal fungi present in microsites with high, intermediate, and low degradation. Different letters indicate significant differences between genera for each microsite ($p < 0.05$). Means and standard error are shown.

	Degradation of the Microsite		
	High	Intermediate	Low
<i>Acaulospora</i> spp.	103.3 $\pm$ 23.0a	101.6 $\pm$ 17.5a	58.2 $\pm$ 12.3a
<i>Ambispora</i> spp.	0.8 $\pm$ 0.5c	0.0 $\pm$ 0.0c	0.2 $\pm$ 0.1c
<i>Claroideoglomus</i> spp.	5.3 $\pm$ 2.8c	6.7 $\pm$ 3.6c	1.0 $\pm$ 0.6c
<i>Entrophospora</i> spp.	0.5 $\pm$ 0.5c	0.0 $\pm$ 0.0c	0.4 $\pm$ 0.3c
<i>Funneliformis</i> spp.	6.5 $\pm$ 2.6c	9.0 $\pm$ 4.1c	7.9 $\pm$ 2.1b
<i>Gigaspora</i> spp.	0.5 $\pm$ 0.2c	0.1 $\pm$ 0.1c	0.2 $\pm$ 0.2c
<i>Glomus</i> spp.	40.8 $\pm$ 13.8b	47.3 $\pm$ 11.3b	27. $\pm$ 6.6b
<i>Scutellospora</i> spp.	0.4 $\pm$ 0.2c	1.2 $\pm$ 0.5c	0.08 $\pm$ 0.08c
<i>Rhizophagus</i> spp.	20.8 $\pm$ 5.9c	12.4 $\pm$ 2.6c	11.00 $\pm$ 3.40c

Table 4. Soil characteristics in the three types of microsites 12.5 years after the beginning of restoration actions. Means and standard deviation ($n = 12$) of soil characteristics are shown. Different letters indicate significant differences between microsites (ANOVA, $p < 0.05$).

Characteristics	Degradation of the Microsite		
	High	Intermediate	Low
Carbon (%)	1.0 $\pm$ 0.3b	4.3 $\pm$ 0.6a	4.6 $\pm$ 0.6a
Organic matter (%)	1.8 $\pm$ 0.4b	7.3 $\pm$ 1.1a	7.9 $\pm$ 1.1a
Nitrogen (%)	0.09 $\pm$ 0.02b	0.4 $\pm$ 0.2a	0.4 $\pm$ 0.2a
C/N ratio	10.96 $\pm$ 0.84a	10.9 $\pm$ 0.5a	11.58 $\pm$ 0.52a
pH in H ₂ O (1:2.5)	5.6 $\pm$ 0.1a	4.9 $\pm$ 0.1b	5.0 $\pm$ 0.1b
Electrical conductivity (mmho/cm)	0.03 $\pm$ 0.01b	0.1 $\pm$ 0.03a	0.1 $\pm$ 0.03a
Phosphorus (ppm)	3.9 $\pm$ 1.2b	10.8 $\pm$ 1.9a	12.9 $\pm$ 3.9a
Nitrates (ppm)	17.2 $\pm$ 5.4b	31.00 $\pm$ 5.68a	69.3 $\pm$ 22.3a
Gravel (%)	11.7 $\pm$ 1.8a	7.7 $\pm$ 2.2a	5.3 $\pm$ 2.3a
Texture classes	Sandy loam	Sandy clay loam	Sandy loam

$p = 0.95$). Neither were any of the three DCA axes correlated with soil characteristics when corrected for multiple tests (27 correlations, all p values > 0.05). The AMF community composition at the morphospecies level as summarized in three DCA axes with eigenvalues of 0.36, 0.18, and 0.08 did not differ significantly between microsite types (ANOVA: axis 1, $F = 0.46$, $p = 0.63$; axis 2, $F = 0.68$, $p = 0.52$; axis 3, $F = 0.73$, $p = 0.48$). Neither were any of the three DCA axes correlated with soil characteristics (27 correlations, all p values > 0.05).

Soil Characterization

Soils at high degradation microsites had significantly lower values of carbon, organic matter, nitrogen, phosphorus, electrical conductivity, and nitrates, and higher pH values than intermediate and low degradation microsites evaluated 12.5 years after planting. Though the intermediate and low degradation sites did not significantly differ in the above soil properties, in every case the intermediate degradation microsites had values, which tended to be intermediate but closer to the low degradation sites than to the high degradation microsites. The C/N and gravel values did not differ between microsite types. Two texture classes were observed: sandy loam soil for high and low degradation microsites, and sandy clay loam soil for intermediate degradation (Table 4).

Discussion

The recovery of degraded sites in the mountains of Central Argentina is a slow process but we discard our worst-case scenario hypothesis where the forest restoration actions of livestock exclusion and the planting of native trees promote the reduction of bare soil at microsites with intermediate degradation but not at the most degraded microsites. Moreover, we also discard the hypothesis of a severe negative effect of degradation on the AMF community. We thus propose that livestock exclusion and the planting of native trees may initiate the reduction of bare soil cover in most situations outside of erosion ravines even in a seasonally dry mountain environment.

Our results showed a reduction in bare soil of 36–37% for the high and intermediate degradation microsites, which represent a rate of vegetation recovery of 2.8% per year. This bare soil reduction is slightly higher than the 2% per year reported in the same study region after livestock exclusion only (Cingolani et al. 2013). The influence of tree planting could be reducing the harsh establishment conditions of bare soils by reducing the impact of raindrops, shading of the soil surface, and provision of organic matter from the tree canopy (Morgan 2005). Reports for other mountain areas without a dry season suggest a much higher reduction in bare soil after livestock exclusion, for example, 17% per year in the montane grasslands of the Southern Pampas of Argentina (Loydi et al. 2012). We

assume that this slower recovery or plant cover in our study region, as compared to the montane grasslands of the Pampas, is due to the long dry season, which slows plant growth when all soil is lost and only bare rock is left (Cingolani et al. 2013) or in more arid regions (i.e. Chartier & Rostagno 2006). In the low degradation microsites, the bare soil was around of 2% and did not significantly change during the 12.5-year period suggesting that those values could be considered satisfactory for the area and a good reference value to attain in future efforts of soil cover restoration.

Mycorrhizal symbiosis is important for plants growing in nutrient-deficient soils (Johnson et al. 2010) and has an important role in ecological restoration (Asmelash et al. 2016; Koziol et al. 2018). In this work, the percentage of AMC was similar in the three microsite types; therefore, the results do not support our hypothesis that AMF community associated with planted *Polylepis australis* would be negatively affected by the characteristics of the high degradation microsites even when these sites had remained without vegetation cover for many years. Thus, our results agree with those of Menoyo et al. (2009) and Soteras et al. (2015) who observed that AMC in *P. australis* forests, where vegetation cover had not been lost, did not differ between different disturbance types such as high livestock density and reduced forest structural complexity, respectively. Mycorrhizal associations are complex systems (Smith & Read 2008) and soil disturbance can alter mycorrhizal colonization through dilution of, or damage to, infective propagules (Jasper et al. 1991; Mickan et al. 2017). In our work, although microsite degradation was heterogeneous, they did not affect AMC in *P. australis*, except for a higher number of hyphal entry points at the low degradation microsites. The mycorrhizal fungal networks play a key role in natural ecosystems (van der Heijden & Horton 2009), and are only connected to roots by entry points that initiate the mycorrhizal associations (Brundrett et al. 1996). The disturbance of soils could result in the disruption of this hyphal network, which can result in the failure of plants to establish (Jasper et al. 1988). So, the high number of entry points observed in low degradation microsites are probably produced good indicators of a healthy mycelium network interacting with plant roots (Klironomos & Hart 2002).

The number of AMF spores was higher at the high and intermediate degradation microsites than at the low degradation site. These results do not support our hypothesis and are consistent with findings reported by Menoyo et al. (2009) who found a higher number of spores at disturbed sites (high livestock density). Abundance and distribution of AMF in natural ecosystems are usually affected by different types of disturbances (Abbott & Robson 1991; Soka & Ritchie 2014; Asmelash et al. 2016), impacting negatively on soil structure through physical disruption (Violi et al. 2008). Soils with good moisture and fertility conditions, such as that in low degradation microsites, reduce AMF sporulation (Oehl et al. 2003; Zandavalli et al. 2008). In this work, the same percentage of AMC was observed among microsite types, but spore number was higher at the high and intermediate degradation sites than at the low degradation microsites. The spore population does not necessarily reflect the contribution to mycorrhizal colonization (Zhao et al. 2001; Liu

& Wang 2003). Soil disturbance may not greatly disturb their ability to colonize roots, such as this study, where spores were not negatively affected by soil disturbance (Hart & Reader 2004) or have resilience to disturbance (Barto et al. 2010; Dai et al. 2013).

A total of 33 AMF morphospecies were identified associated with *P. australis*, which is consistent with the 32 morphospecies detected by Soteras et al. (2015) in *P. australis* forests in the Córdoba Mountains. This result shows some specificity in AMF communities with the *P. australis*-dominated community. The genera *Acaulospora* and *Glomus* were the most abundant, in agreement with previous findings (e.g. Qing-Ming & Liang-Dong 2010; Stürmer & Siqueira 2011; Chagnon et al. 2013; Soteras et al. 2015).

Different AMF taxa can differ in their symbiotic function (Helgason et al. 2007; Lee et al. 2013), which is beneficial for a plant if colonized by different species that are complementary in their functions, such as nutrient uptake from different sources in the soil (Maherali & Klironomos 2007), especially in environments with barely favorable abiotic conditions (Chagnon et al. 2013). The highest AMF species richness was observed at the high degradation sites although there were no differences in the diversity index; thus, our hypothesis is once again rejected. Higher AMF richness at high degradation microsites compared to the other microsites could be related to the reduction of bare soil after a period of 12.5 years since the restoration of the montane forest using a native species. Apparently, in disturbed ecosystems, some AMF morphospecies that are adapted to specific local conditions tend to be permanently lost (Brundrett 1991), but in our case study the diversity was not different. AMF can survive, adapt, and disperse under changing soil conditions during succession (Picone 2000; Zobel & Öpik 2014; Soteras et al. 2015), which might explain the richness and diversity values found in this community.

We conclude that livestock exclusion and the planting of native trees may initiate the reduction of bare soil cover in most situations out of erosion ravines, and that the AMF community is resilient to soil degradation. As our study was performed away from erosion gullies, our conclusions may not apply to these sites where active soil loss is severe and active restoration techniques should probably be applied.

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LITERATURE CITED

- Abbott LK, Robson AD (1991) Factors influencing the occurrence of vesicular-arbuscular mycorrhizas. *Agriculture, Ecosystems and Environment* 35:121–150
- Asmelash F, Bekele T, Birhane E (2016) The potential role of arbuscular mycorrhizal fungi in the restoration of degraded lands. *Frontiers in Microbiology* 7:1095

- Augé RM (2001) Water relations, drought and vesicular mycorrhizal symbiosis. *Mycorrhiza* 11:3–42
- Barto EK, Alt F, Oelmann Y, Wilcke W, Rillig MC (2010) Contributions of biotic and abiotic factors to soil aggregation across a land use gradient. *Soil Biology and Biochemistry* 42:2316–2324
- Bellis LM, Pidgeon AM, Alcantara C, Dardanelli S, Radeloff VC (2015) Influences of succession and erosion on bird communities in a South American highland wooded landscape. *Forest Ecology and Management* 349:85–93
- Błaszowski J (2013) Taxonomy of arbuscular fungi: species descriptions and illustrations. <http://www.zor.zut.edu.pl/Glomeromycota/Taxonomy.html> (accessed 10 Jul 2018)
- Bower CA, Wilcox LW (1965) Soluble salts. Pages 933–951. In: Black CA (ed) *Methods in soil analysis: agronomy*. American Society Agronomy Inc., Madison, Wisconsin
- Bremner JM, Mulvaney CS (1982) Nitrogen-total. Pages 595–624. In: Page AL, Miller RH, Keeney DR (eds) *Methods of soil analysis II*. American Society Agronomy Inc., Madison, Wisconsin
- Brundrett M (1991) Mycorrhizas in natural ecosystems. *Advances in Ecological Research* 21:171–262
- Brundrett M, Bougher N, Dell B, Grove T, Malajczuk N (1996) Working with mycorrhizas in forestry and agriculture. *ACIAR Monograph* 32, Australian Centre for International Agricultural Research, Canberra, Australia
- Cabido M (1985) Las comunidades vegetales de la Pampa de Achala, Sierras de Córdoba, Argentina. *Documents Phytosociologiques* 9:431–443
- Cabido M, Breimer R, Vega G (1987) Plant communities and associated soil types in a high plateau of the Cordoba Mountains, Central Argentina. *Mountain Research and Development* 7:25–42
- Chagnon PL, Bradley RL, Maherali H, Klironomos JN (2013) A trait-based framework to understand life history of mycorrhizal fungi. *Trends in Plant Science* 18:484–491
- Chartier MP, Rostagno CM (2006) Soil erosion thresholds and alternative states in northeastern Patagonian rangelands. *Rangeland Ecology & Management* 59:616–624
- Cingolani AM, Renison D, Zak MR, Cabido MR (2004) Mapping vegetation in a heterogeneous mountain rangeland using landsat data: an alternative method to define and classify land-cover units. *Remote Sensing of Environment* 92:84–97
- Cingolani AM, Vaieretti MV, Giorgis MA, La Torre N, Whitworth-Hulse JI, Renison D (2013) Can livestock and fires convert the sub-tropical mountain rangelands of central Argentina into a rocky desert? *The Rangeland Journal* 35:285–297
- Cingolani AM, Vaieretti MV, Giorgis MA, Poca M, Tecco PA, Gurvich DE (2014) Can livestock grazing maintain landscape diversity and stability in an ecosystem that evolved with wild herbivores? *Perspectives in Plant Ecology, Evolution and Systematics* 16:143–153
- Clark RB, Zeto SK (2000) Mineral acquisition by arbuscular mycorrhizal plants. *Journal of Plant Nutrition* 23:867–902
- Colladon L, Felici GS, Pazos I (2010) Anuario pluviométrico 2005–2010. Cuenca del Río San Antonio. Sistema del Río Suquia – Provincia de Córdoba. Instituto Nacional del Agua y del Ambiente y Centro de Investigación de la Región Semiárida (CIRSA), Córdoba, Argentina
- Dai ML, Bainard LD, Hamel C, Gan YT, Lynch D (2013) Impact of land use on arbuscular mycorrhizal fungal communities in rural Canada. *Applied and Environmental Microbiology* 79:6719–6729
- Doetterl S, Berhe AA, Nadeu E, Wang Z, Sommer M, Fiener P (2016) Erosion, deposition and soil carbon: a review on process-level controls, experimental tools and models to address C cycling in dynamic landscapes. *Earth-Science Reviews* 154:102–122
- Durán Zuazo VH, Rodríguez Pleguezuelo CR (2008) Soil-erosion and runoff prevention by plant covers. A review. *Agronomy for Sustainable Development* 28:65–86
- Fajardo L, Cuenca G, Arrindell P, Capote R, Hasmy Z (2011) El uso de los Hongos micorrizicos arbusculares en las prácticas de restauración ecológica. *Interciencia* 36:931–936
- FAO (Food and Agriculture Organization of the United Nations) (1993) Erosión de suelos en América Latina. <http://www.fao.org/docrep/t2351s/T2351S00.htm> (accessed 22 Mar 2017)
- FAO (2015) Los bosques y suelos forestales contribuyen de manera esencial a la producción agrícola y la seguridad alimentaria mundial. <http://www.fao.org/soils-2015/news/news-detail/es/c/285875> (accessed 10 May 2017)
- Gerdemann JW, Nicolson TH (1963) Spores of mycorrhizal endogone species extracted from soil by wet sieving and decanting. *Transactions of the British Mycological Society* 46:235–244
- Grace C, Stribley CP (1991) A safer procedure for routine staining of vesicular arbuscular mycorrhizal fungi. *Mycological Research* 95:1160–1162
- Green K, Good R, Johnston S, Simpson L (2005) Alpine grazing in the Snowy Mountains of Australia: degradation and stabilization of the ecosystem. Pages 213–225. In: Spehn E, Liberman M, Körner C (eds) *Land use changes and mountain biodiversity*. CRC Press LLC, Boca Raton, Florida
- Hart M, Reader R (2004) Do arbuscular mycorrhizal fungi recover from soil disturbance differently? *Tropical Ecology* 45:97–111
- Helgason T, Merryweather JW, Young JPW, Fitter AH (2007) Specificity and resilience in the arbuscular mycorrhizal fungi of a natural woodland community. *Journal of Ecology* 95:623–630
- Hijmans RJ, Cameron SE, Parra JL, Jones PG, Jarvis A (2005) Very high resolution interpolated climate surfaces for global land areas. *International Journal of Climatology* 25:1965–1978
- InfoStat (2017) Grupo InfoStat, FCA. Universidad Nacional de Córdoba, Córdoba, Argentina. <http://www.infostat.com.ar>
- INVAM (International Vesicular Arbuscular Mycorrhiza) (2018) International collection of vesicular-arbuscular mycorrhizal fungi. <http://invam.caf.wvu.edu/fungi/taxonomy/speciesID.htm> (accessed 10 Jul 2018)
- Jackson ML (1964) Análisis químico de suelos. Omega, Barcelona, España
- Jasper DA, Robson AD, Abbott LK (1988) Revegetation in an iron-ore mine nutrient requirements for plant growth and the potential role of vesicular arbuscular (VA) mycorrhizal fungi. *Australian Journal of Soil Research* 26:497–507
- Jasper DA, Abbott LK, Robson AD (1991) The effect of soil disturbance on vesicular-arbuscular mycorrhizal fungi in soils from different vegetation types. *New Phytologist* 118:471–476
- Johnson NC, Wilson GWT, Bowker MA, Wilson JA, Miller RA (2010) Resource limitation is a driver of local adaptation in mycorrhizal symbioses. *Proceedings of the National Academy of Sciences of the United States of America* 107:2093–2098
- Klironomos JN, Hart MM (2002) Colonization of roots by arbuscular mycorrhizal fungi using different sources of inoculum. *Mycorrhiza* 12:181–184
- Körner C (2003) Carbon limitation in trees. *Journal of Ecology* 91:4–17
- Kozioł L, Schultz PA, House GL, Bauer JT, Middleton EL, Bever JD (2018) The plant microbiome and native plant restoration: the example of native mycorrhizal fungi. *Bioscience* 68:996–1006
- Lee E-H, Eo J-K, Ka K-H, Eom A-H (2013) Diversity of arbuscular mycorrhizal fungi and their roles in ecosystems. *Mycobiology* 41:121–125
- Leifheit EF, Veresoglou SD, Lehmann A, Morris EK, Rillig MC (2014) Multiple factors influence the role of arbuscular mycorrhizal fungi in soil aggregation – a meta-analysis. *Plant and Soil* 374:523–537
- Liu RJ, Wang FY (2003) Selection of appropriate host plants used in trap culture of arbuscular mycorrhizal fungi. *Mycorrhiza* 13:123–127
- Loydi A, Zalba SM, Distel RA (2012) Vegetation change in response to grazing exclusion in montane grasslands, Argentina. *Plant Ecology and Evolution* 145:313–322
- Magurran AE (1988) *Ecological diversity and its measurement*. Croom Helm, London, United Kingdom
- Maherali H, Klironomos JN (2007) Influence of phylogeny on fungal community assembly and ecosystem functioning. *Science* 316:1746–1748
- McGonigle TP, Miller MH, Evans DG, Fairchild GL, Swan JA (1990) A method which gives an objective measure of colonization of roots by vesicular-arbuscular mycorrhizal fungi. *New Phytologist* 115:495–501
- Meier IC, Brunner I, Godbold DL, Helmisaarilvika H-S, Ostonen I, Soudzilovskaia NA, Prescott CE (2019) Roots and rhizospheres in

- forest ecosystems: recent advances and future challenges. *Forest Ecology and Management* 431:1–5
- Menoyo E, Renison D, Becerra AG (2009) Arbuscular mycorrhizas and performance of *Polylepis australis* trees in relation to livestock density. *Forest Ecology and Management* 258:2676–2682
- Mickan BS, Abbott LK, Fan J, Hart MM, Siddique KH, Solaiman ZM, Jenkins SN (2017) Application of compost and clay under waterstressed conditions influences functional diversity of rhizosphere bacteria. *Biology and Fertility of Soils* 54:55–70
- Morgan RPC (2005) Soil erosion and conservation. Blackwell Publishing, Oxford, United Kingdom
- Nelson DW, Sommers LE (1982) Total carbon, organic carbon and organic matter. Pages 539–580. In: Page AL, Miller RH, Keeney DR (eds) *Methods of soil analysis II*. American Society Agronomy Inc., Madison, Wisconsin
- Oehl F, Sieverding E, Ineichen K, Mäder P, Boller T, Wiemken A (2003) Impact of land use intensity on the species diversity of arbuscular mycorrhizal fungi in agroecosystems of Central Europe. *Applied and Environmental Microbiology* 69:2816–2824
- Omar MB, Bolland L, Heather WA (1979) P.V.A. (polyvinyl alcohol). A permanent mounting medium for fungi. *Bulletin of the British Mycological Society* 13:31–32
- Peech M (1965) Hydrogen ion activity. Pages 914–926. In: Black CA (ed) *Methods in soil analysis: agronomy*. American Society Agronomy Inc, Madison, Wisconsin
- Phillips JM, Hayman DS (1970) Improved procedures for clearing roots and staining parasitic and vesicular-arbuscular mycorrhizal fungi for rapid assessment of infection. *Transactions of the British Mycological Society* 55:158–161
- Picone C (2000) Diversity and abundance of arbuscular mycorrhizal fungi spores in tropical forest and pasture. *Biotropica* 32:734–750
- Pimentel D, Harvey C, Resosudarmo P, Sinclair K, Kurz D, McNair M, et al. (1995) Environmental and economic costs of soil erosion and conservation benefits. *Science* 267:1117–1123
- Qing-Ming G, Liang-Dong G (2010) A comparative study of arbuscular mycorrhizal fungi in forest, grassland and cropland in the Tibetan Plateau, China. *Mycology* 1:163–170
- Redecker D, Schussler A, Stockinger H, Stürmer SL, Morton JB, Walker C (2013) An evidence-based consensus for the classification of arbuscular mycorrhizal fungi (Glomeromycota). *Mycorrhiza* 23:515–531
- Renison D, Cingolani AM, Suárez R, Menoyo E, Coutiers C, Sobral A, Hensen I (2005) The restoration of degraded mountain woodlands: effects of seed provenance and microsite characteristics of *Polylepis australis* seedling survival and growth in Central Argentina. *Restoration Ecology* 13:129–137
- Renison D, Hensen I, Suarez R (2011) Landscape structural complexity of high-mountain *Polylepis australis* forests: a new aspect of restoration goals. *Restoration Ecology* 19:390–398
- Renison D, Herrero ML, Torres RC, Suarez R, Friedlander P, Navarro Ramos SE, Barri F, Cingolani AM (2016) El rol de los voluntariados en la restauración ecológica del centro argentino. Pages 55–171. In: Manzini Vázquez (ed) *Más allá de la ecología de la restauración: perspectivas sociales en América Latina y el Caribe*. 1st ed. SIACRE, Buenos Aires, Argentina
- Robledo GL, Renison D (2010) Wood-decaying polypores in the mountains of central Argentina in relation to *Polylepis* forest structure and altitude. *Fungal Ecology* 3:178–184
- Schenck NC, Pérez Y (eds) (1987) *Manual for identification of VA mycorrhizal fungi*. INVAM, University of Florida, Gainesville
- Sikes BA (2010) When do arbuscular mycorrhizal fungi protect plant roots from pathogens? *Plant Signaling and Behaviour* 5:763–765
- Simoes Macayo N, Renison D (2015) Cuántos años monitorear el éxito de plantaciones con fines de restauración?: Análisis en relación al micrositio y procedencia de las semillas. *Bosque* 36:315–322
- Smith SE, Read DJ (2008) *Mycorrhizal symbiosis*. Academic Press, San Diego, California
- Smith MD, Hartnett DC, Rice CW (2000) Effects of long-term fungicide applications on microbial properties in tall grass prairie soil. *Soil Biology and Biochemistry* 32:935–946
- Soil Survey Staff (2010) *Keys to soil taxonomy*. 10th edition. United States Department of Agriculture, Natural Resources Conservation Service, Washington D.C.
- Soka G, Ritchie M (2014) Arbuscular mycorrhizal symbiosis and ecosystem processes: prospects for future research in tropical soils. *Journal of Ecology* 4:11–22
- Soteras F, Renison D, Becerra AG (2013) Growth response, phosphorus content and root colonization of *Polylepis australis* Bitt. seedlings inoculated with different soil types. *New Forest* 44:577–589
- Soteras F, Grilli G, Cofré MN, Marro N, Becerra A (2015) Arbuscular mycorrhizal fungal composition in high montane forests with different disturbance histories in central Argentina. *Applied Soil Ecology* 85:30–37
- Stürmer SL, Siqueira JO (2011) Species richness and spore abundance of arbuscular mycorrhizal fungi across distinct land uses in western Brazilian Amazon. *Mycorrhiza* 21:255–267
- Ter Braak CJF (1987) *CANOCO — a FORTRAN program for canonical community ordination*. Microcomputer Power, Ithaca, New York
- van der Heijden MGA, Horton TR (2009) Socialism in soil? The importance of mycorrhizal fungal networks for facilitation in natural ecosystems. *Journal of Ecology* 97:1139–1150
- Violi HA, Barrientos-Priego AF, Wright SF, Escamilla-Prado E, Morton JB, Menge JA, Lovatt CJ (2008) Disturbance changes arbuscular mycorrhizal fungal phenology and soil glomalin concentrations but not fungal spore composition in montane rainforests in Veracruz and Chiapas, Mexico. *Forest Ecology and Management* 254:276–290
- Walker C, Mize W, McNabb HS (1982) Populations of endogoneaceous fungi at two populations in central Iowa. *Canadian Journal of Botany* 60:2518–2529
- Wang B, Qiu YL (2006) Phylogenetic distribution and evolution of mycorrhizas in land plants. *Mycorrhiza* 16:299–363
- Zak MR, Cabido M, Hodgson JG (2004) Do subtropical seasonal forests in the Gran Chaco, Argentina, have a future? *Biological Conservation* 120:589–598
- Zandavalli RB, Stürmer SL, Dillenburg LR (2008) Species richness of arbuscular mycorrhizal fungi in forest with *Araucaria* in Southern Brazil. *Hoehnea* 35:63–68
- Zhao ZW, Xia YM, Qin XZ, Li XW, Chen LZ, Sha T, Wang GH (2001) Arbuscular mycorrhizal status of plants and the spore density of arbuscular mycorrhizal fungi in tropical rain forest of Xishuangbanna, southwest China. *Mycorrhiza* 11:159–162
- Zobel M, Öpik M (2014) Plant and arbuscular mycorrhizal fungal (AMF) communities – which drives which? *Journal of Vegetation Science* 25:1133–1140

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